Gen-AI for Academic Writing/ Research

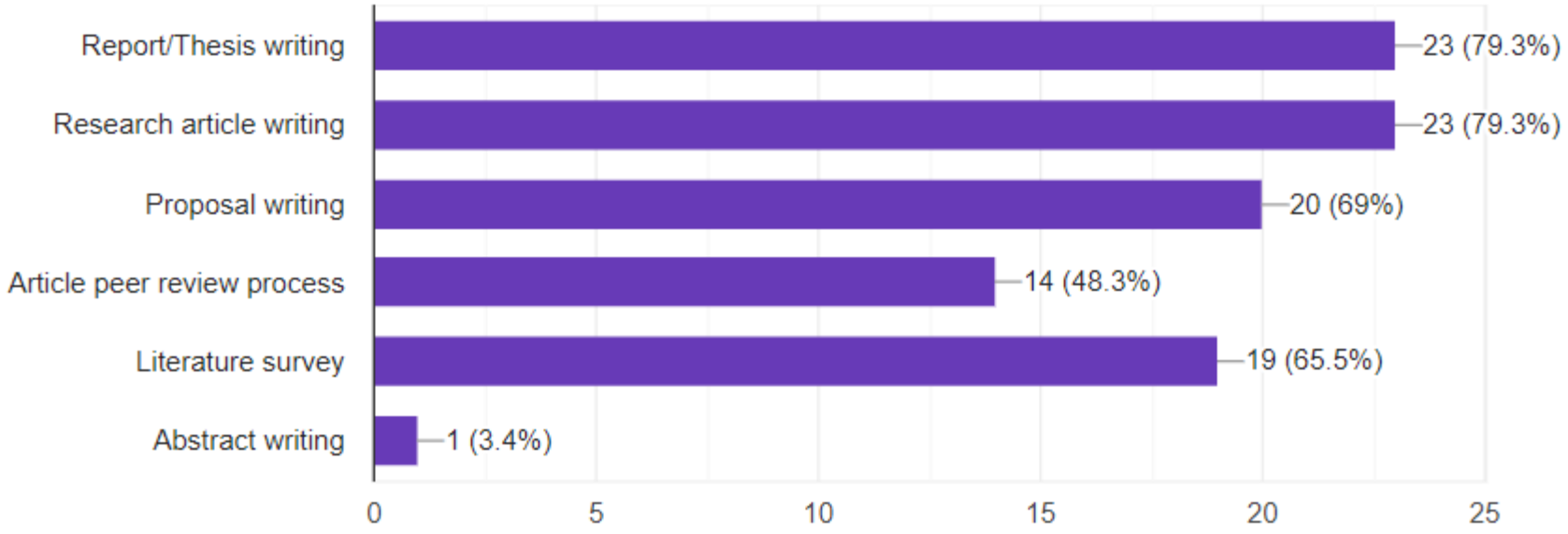
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Outcomes..

- Have gained practical experience in utilizing AI tools and techniques **to enhance your academic writing skills**.
- Be equipped with **strategies to optimize each section of your research paper** following the **IMRD** (Introduction, Methods, Results, Discussion) framework.
- Use the **ethical considerations** surrounding AI-powered academic writing and be able to navigate them responsibly.

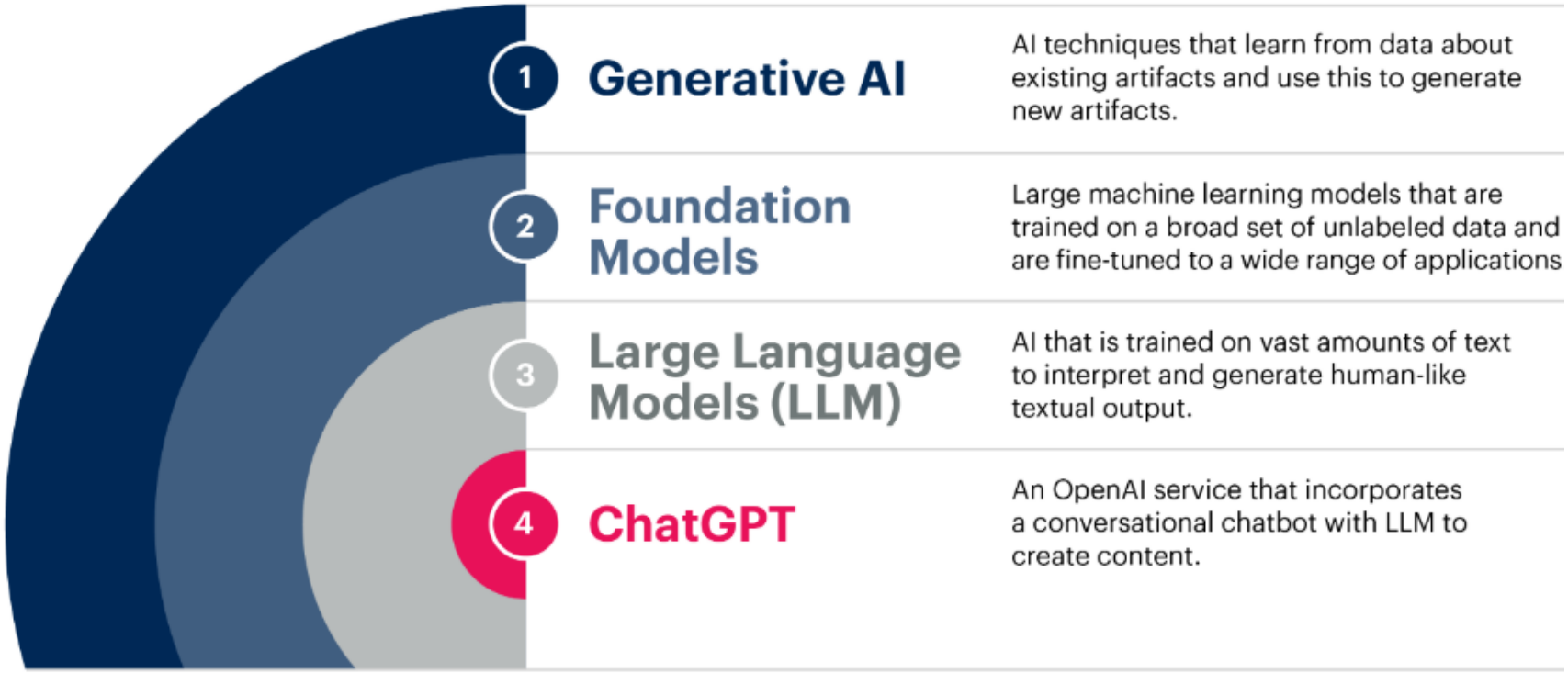
Extracts from Survey Results..



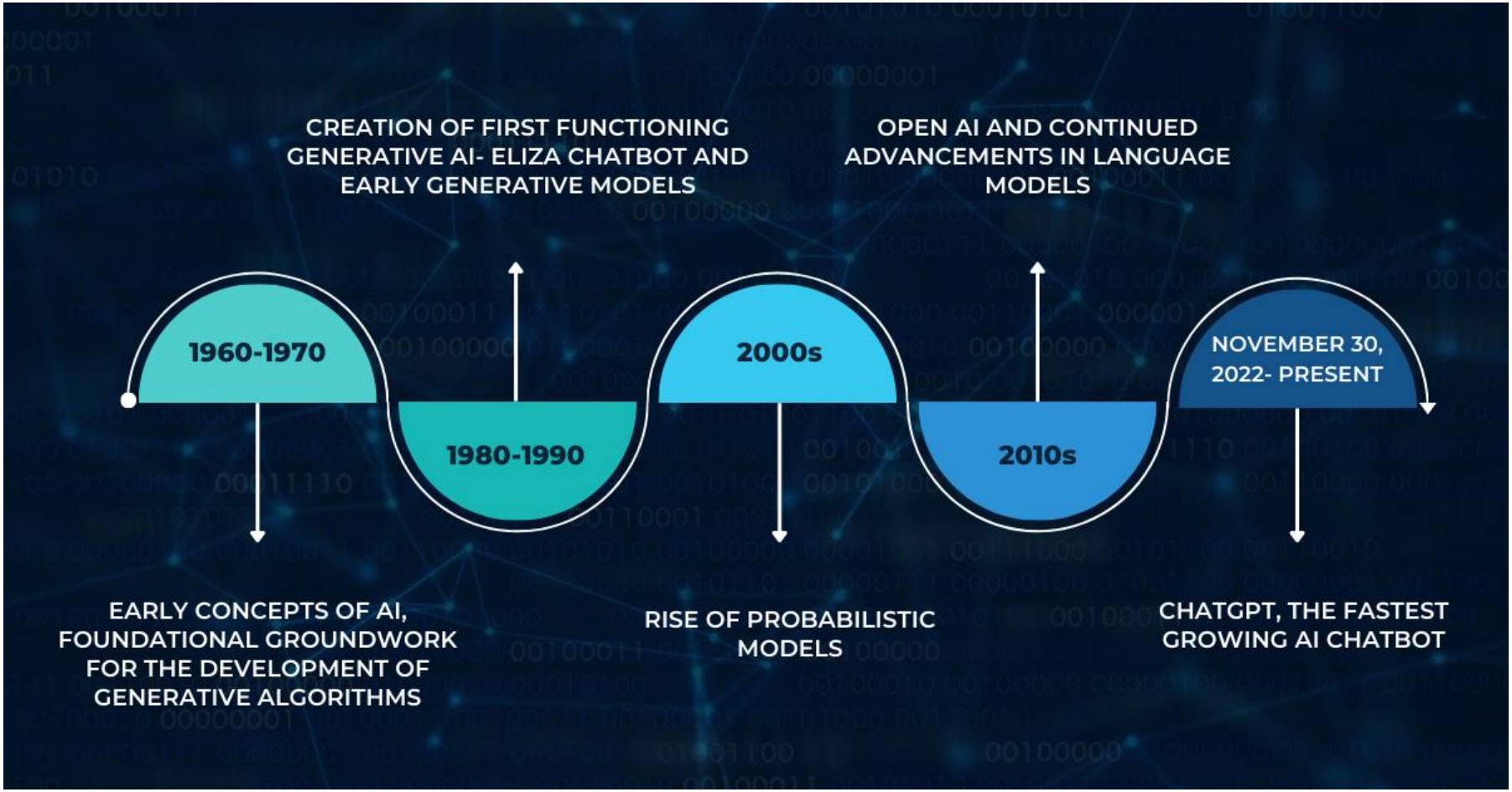
Outline

- Functionalities of tools to enhance academic writing (eg: ChatGPT/ Gemnai/Copilit)
- How to use AI writing tools in idea generation, drafting, editing in academic writing (specially in research article writing)?

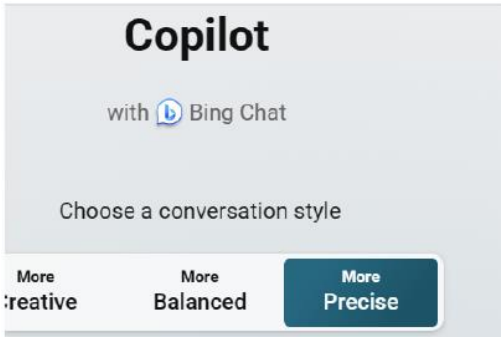
Generative-AI



Generative-AI

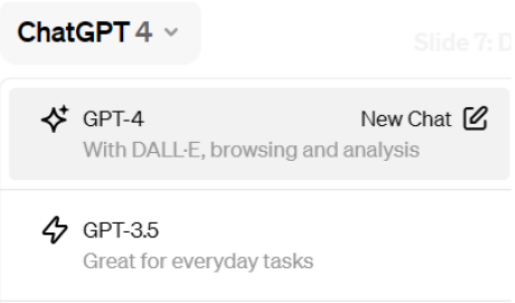


Generative-AI



What is Gen-AI?

- A broad label that encompasses any type of AI that can GENERATE new content text, audio, video, graphics etc.



What is the difference between Gen-AI and Normal AI?

- Gen-AI: Generates **NEW** content
- Normal AI/ Traditional AI/ Discriminative AI: focuses on identifying distinct patterns, decision making and classification

Types of Generative-AI

Purpose of G-AI Tools	Types of Tools (some examples)
Text	GPT, Jasper, AI-Writer and Lex
Image	Dall-E 2, Midjourney and Stable Diffusion
Music	Amper, Dadabots and MuseNet
Code	CodeStarter, Codex, GitHub Copilot and Tabnine
Voice	Descript, Listnr and Podcast.ai
Video	Steve.ai, and Designs.ai

How does Gen-AI work?

"Don't cry ....."

"Don't cry over ....."

Data Learning	• It starts by absorbing large datasets to learn patterns and structures.
Pattern Recognition	• Utilizes algorithms to analyse and understand these patterns.
Content Generation	• After learning, it can create new, similar yet original outputs.
Feedback-Based Refinement	• AI refines its outputs by learning from evaluations and adjustments.
Continuous Adaptation	• AI evolves, improving its output quality and relevance.

Gen-AI- Basic Functions

- Relevant
- Accurate
- Coherent

Quality of Response



- Clarity
- Specificity
- Relevance

Quality of Prompt

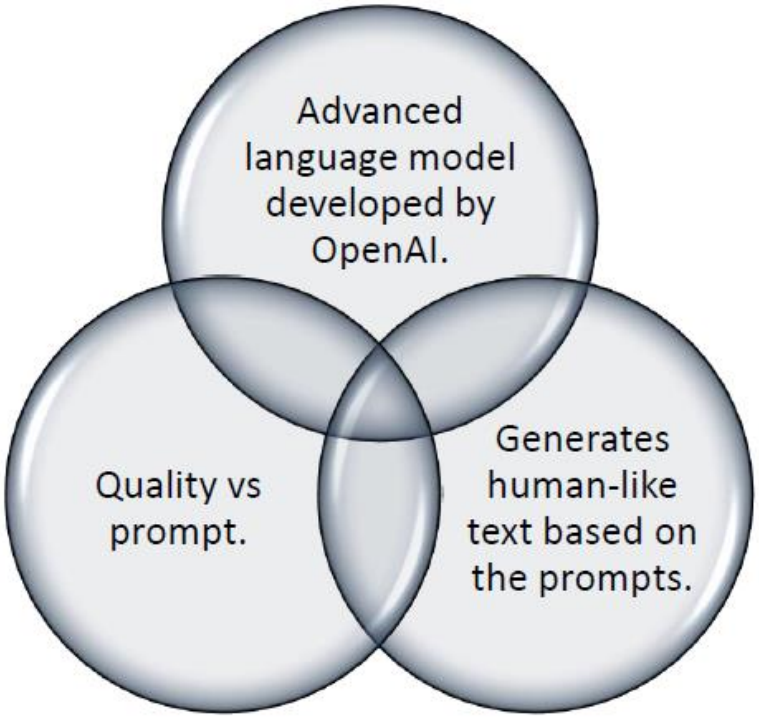


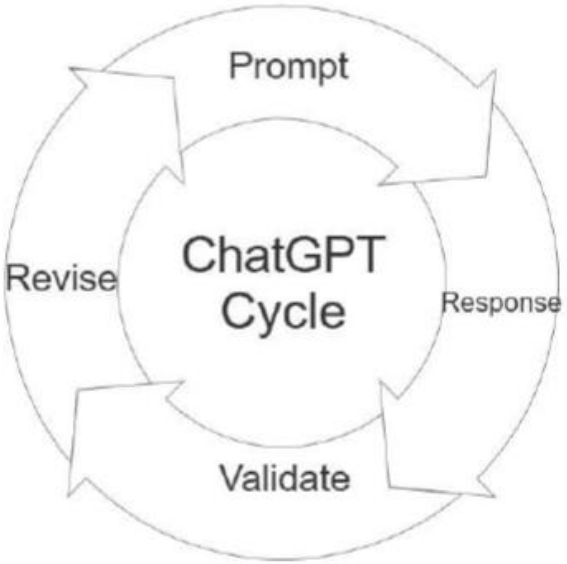
- Ability to interpret
- Training
- Design
- Current context.

Model Understanding



Understanding Gen-AI





Quality of Response ∝ Quality of Prompt × Model Understanding

Quality of Response is the measure of how relevant, accurate, and coherent the response is.

Quality of Prompt represents the clarity, specificity, and relevance of the prompt given to the model.

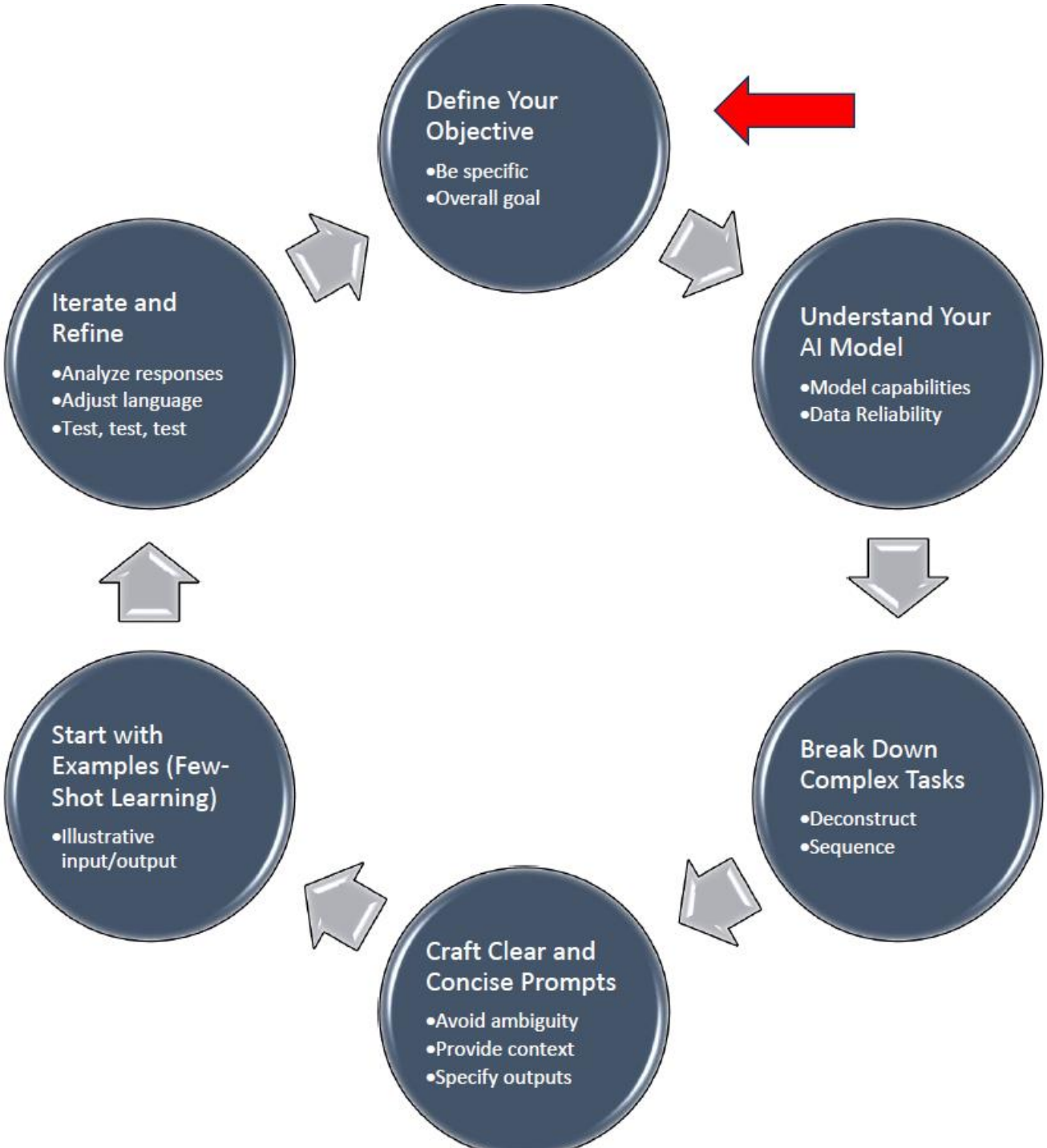
Model Understanding , model's ability to interpret the prompt, including its training, design, and current context.

Prompting

- Refers to the request/question or input that you provide to the model
- The model takes this prompt and generates a response based on the information it has been trained on.

- Initial Statement or Question
- Context
- Intended Output
- Tone or Formality
- Specificity
- Instructions for Response Format

Prompt Generation Steps



Prompt Generation and Response Validation (Important Points to remember)

Define the Objective:	• Identify the specific information or assistance
Be Clear and Precise:	• Use clear language and avoid ambiguity. • Include essential details without over-complicating the prompt.
Consider Context:	• Provide relevant background or context to guide the response.
Set the Tone and Style:	• Specify the desired tone (formal, casual) or style (e.g., summary, explanation) if it matters for your use case.
Ask Direct Questions:	• If seeking specific information, formulate your prompt as a direct question. • Self-Reflective
Avoid Bias and Leading Questions:	• Craft the prompt neutrally to prevent biased or skewed responses.
Test and Refine:	• Experiment with different phrasings and observe how slight changes can affect the response. • Refine the prompt
Consider Ethical and Privacy Concerns:	• Ethical guidelines and does not request or reveal sensitive or private information.

Good Prompt vs Bad Prompt

Field	Good Prompt	Bad Prompt
Economy	Generate a comprehensive analysis of the impact of digital currencies on traditional banking systems within Southeast Asia since 2018, focusing on consumer behavior, regulatory responses, and implications for financial stability.	Write something about the economy.
Education	Develop a structured review of the literature on the effectiveness of blended learning models in higher education for promoting student engagement and academic achievement, including an assessment of comparative studies from the last decade.	What's new in education?
Computer Science	Outline the evolution of machine learning algorithms for natural language processing (NLP) from 2010 to 2023, highlighting key technological advancements, their impact on AI-driven applications, and emerging trends in algorithm efficiency and accuracy.	Explain computer science.

Response Validation (RI meets AI)

Exercise 1 (Hands-on):

Type the following on ChatGPT:

How frequent is the cell reproduction is done in humans?

Check the output

Provide the references to the above information.

Go through every reference it provides. Are they all reliable? Can you validate every response?

Prompt Engineering

- The primary goal is **ability to achieve optimized outputs with minimal post-generation effort**.
- Generative AI outputs can be mixed in quality, often requiring skilled practitioners to review and revise.
- By crafting precise prompts, prompt engineers ensure that AI-generated output aligns with the desired goals and criteria, **reducing the need for extensive post-processing.**

Response Validation (RI meets AI)

Exercise 2 (Hands-on):

"How frequent is the cell reproduction is done in humans?"
"Provide the references to the above information."

Refine the previous prompt/ write a new one to receive better output and post them on padlet (with more references to referred/indexed journals/conferences)

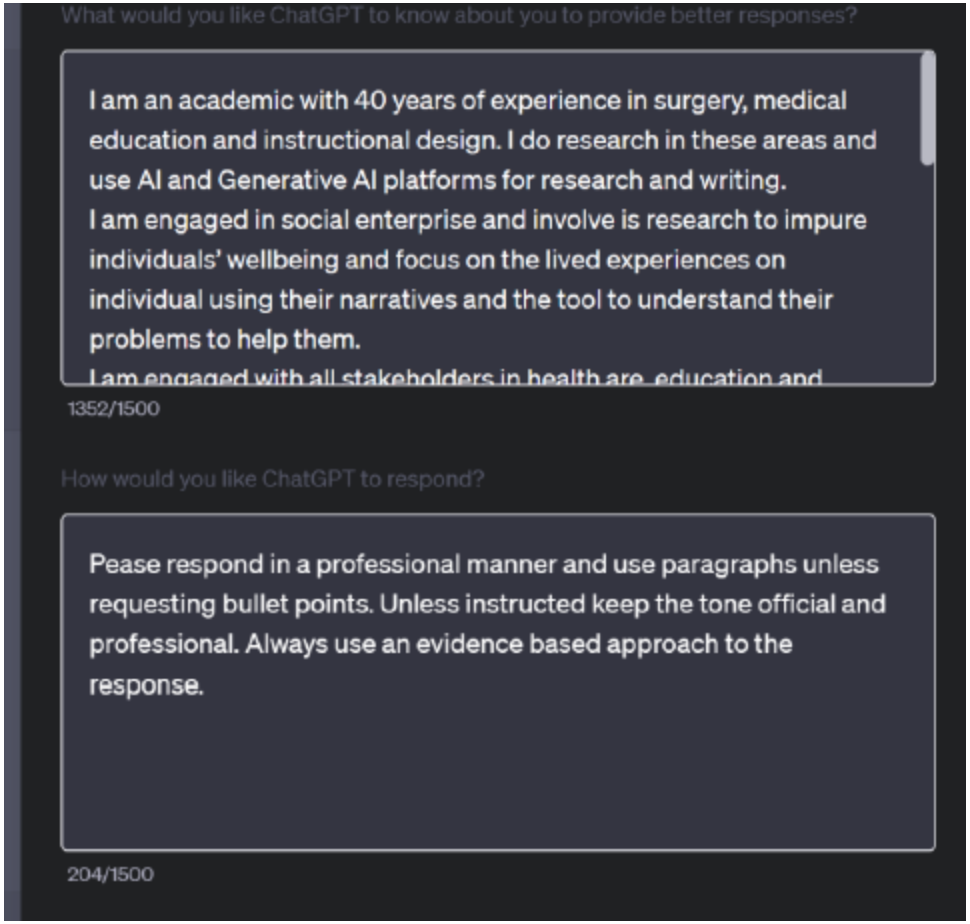
https://padlet.com/smv29/genai-response-validation-yjyjlxrzyuvm3a52

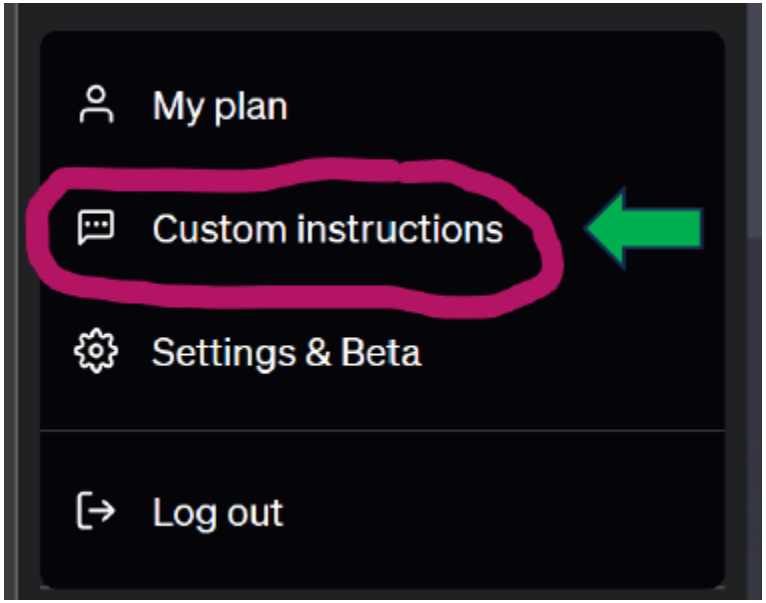
Sample Answer

Provide how frequent is the cell reproduction is done in humans by referring to the peer-reviewed journal articles

- Validate with the links to the given references
- Paid version of ChatGPT→ provides the recent references (ChatGPT 3.5 → provides up to 2021)
- Can use other tools for this (eg: Elicit, Typeset, with zotero/ Mendeley – a reference management s/w)

Personalizing Gen-AI

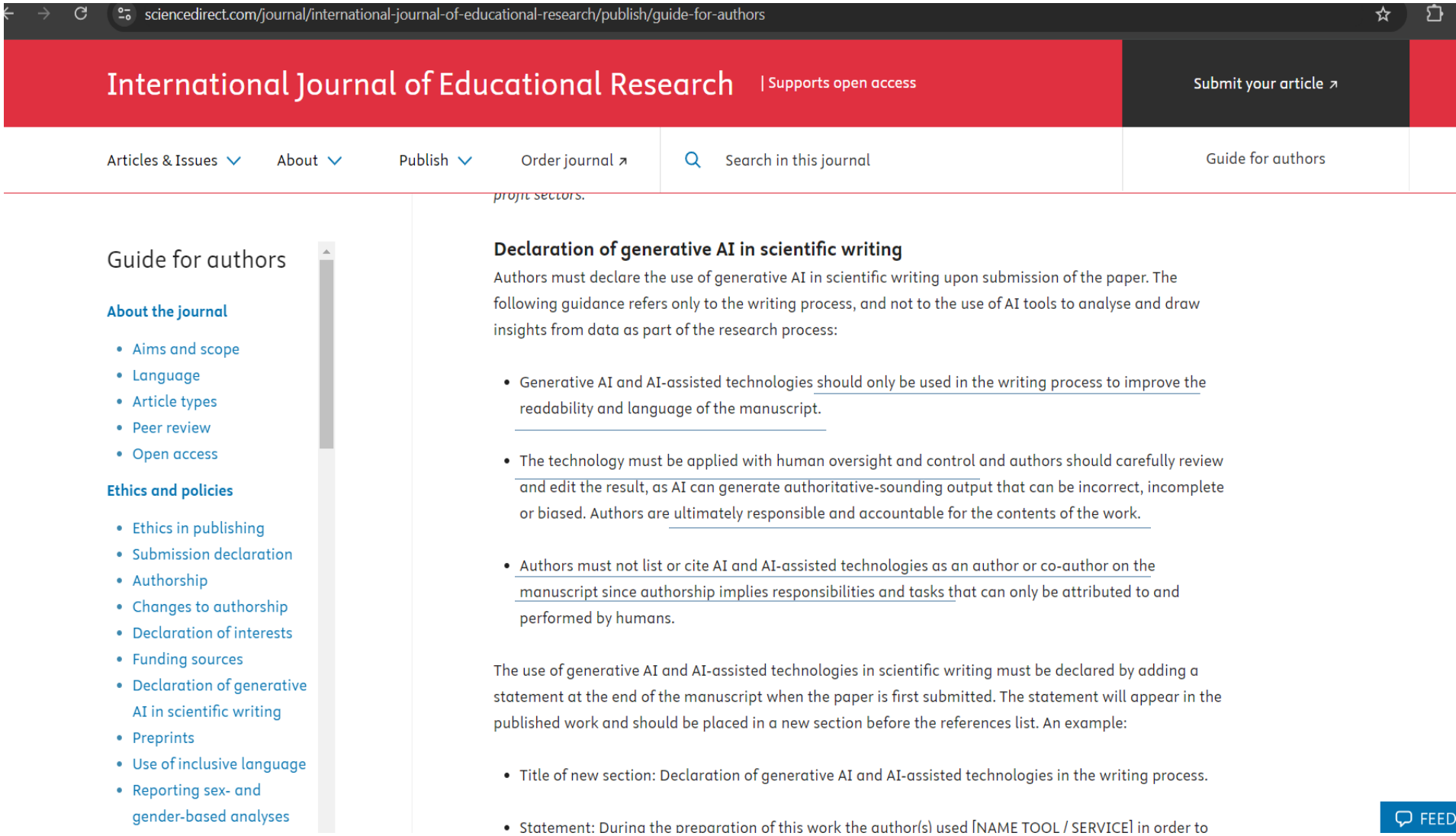




Feature / Model	GPT-4	Gemini	Copilot
Primary Use	Broad spectrum text generation	Conversational AI and information retrieval	Enhanced search experiences with AI integration
Training Data	Diverse text from books, websites, etc.	Likely integrated with Google's search data	Integrates Bing search data with AI capabilities
Strengths	Versatile text generation, deep contextual understanding, wide application range	Integration with Google's search capabilities, conversational responses	Enhanced search results, interactive and context-aware responses
Output Type	Text across various domains and styles	Informational and conversational responses	Enhanced and interactive search query responses
Interactivity	High, with dynamic and contextually relevant responses	Likely high, focused on conversational engagement	High, especially in enhancing search interactions
Real-time Data Access	Limited to training data up to 2023	Likely access to real-time data from Google	Access to real-time data through Bing search engine
Application Scope	Wide-ranging across industries and sectors	Mainly for information retrieval and conversational AI	Primarily enhancing search experiences and providing detailed search responses
Customization and Flexibility	High, adaptable to various tasks and industries	Focused on conversational AI, flexibility in response style	Integrates with search engine capabilities, adaptable in responding to varied search queries

Gen-AI for Article Writing

Declaration of Gen-AI in Journals



GenAI Policies for Journals

Generative AI policies for journals

About For authors For reviewers For editors Frequently asked questions

These policies have been triggered by the rise of generative AI* and AI-assisted technologies, which are expected to increasingly be used by content creators. These policies aim to provide greater transparency and guidance to authors, reviewers, editors, readers and contributors. Elsevier will monitor this development and will adjust or refine policies when appropriate.

For authors

The use of generative AI and AI-assisted technologies in scientific writing

Gen-AI Support for Article Writing

- **Exercise 3 (Hands-on):**

Craft a research question that you would like to explore using various Gen-AI and post the responses on padlet.

- ChatGPT/ Gemini/ CoPilot

- https://padlet.com/smv29/crafting-a-research-question-42tif4xxaxeha9bk

Ethical Considerations

- Privacy issues when signing up to the tools
- Copyright issues
- Accuracy issues (unreliable sources, hallucination etc.)
  - ChatGPT sometimes fabricates citations to sources that do not exist

Tools to Improve Academic Writing

- 'Elicit' for the **Literature search**
- 'Scholarcy' and 'Typeset' for **Data Extraction and Summary**
- 'Genei.io' for **Summarization and Key Points Highlighting**
- **Keyword Generation** with 'ChatGPT' (targeted prompt engineering)

Hands on Session on Elicit

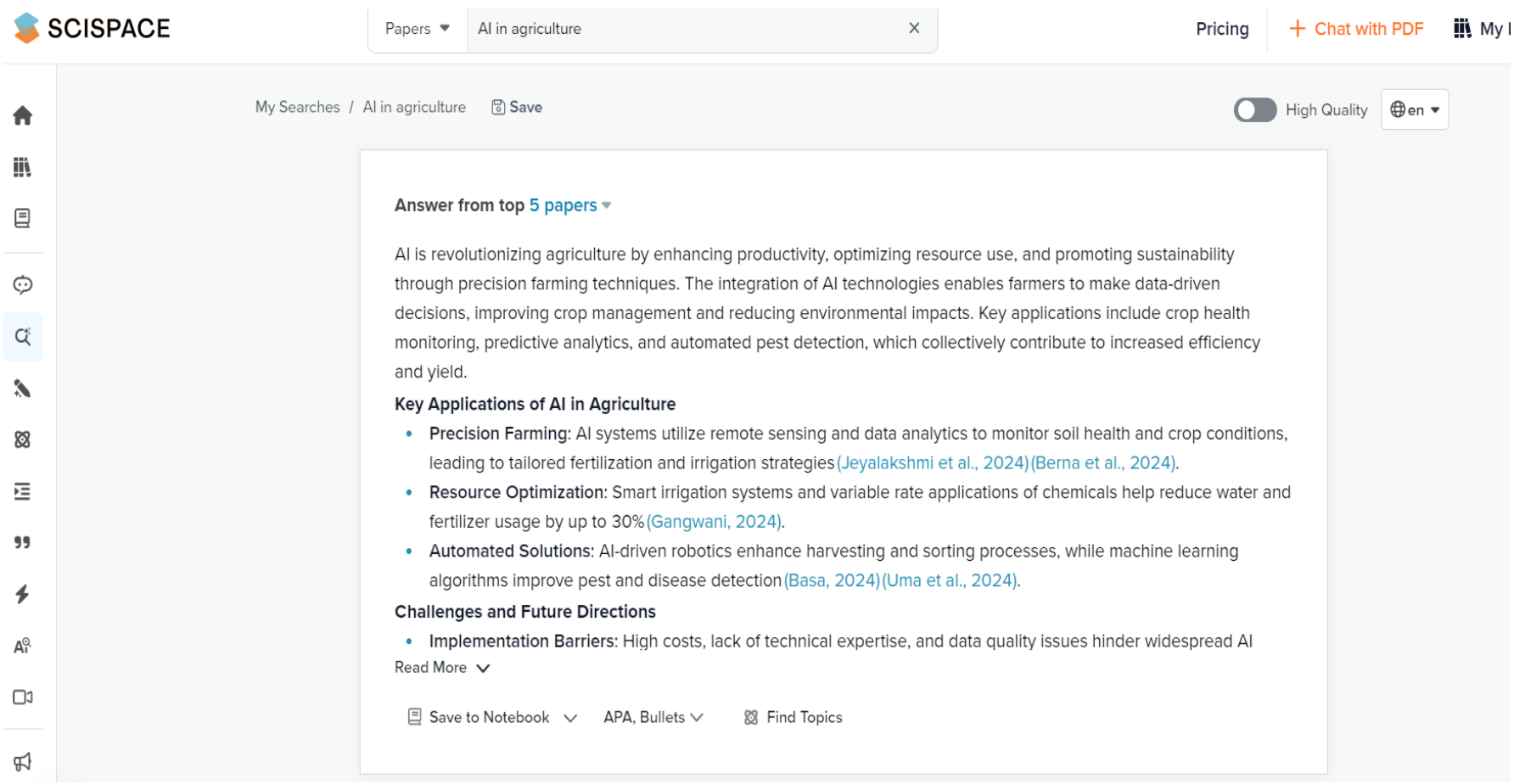
Hands on Session on Typeset (Scispace)


Q
A

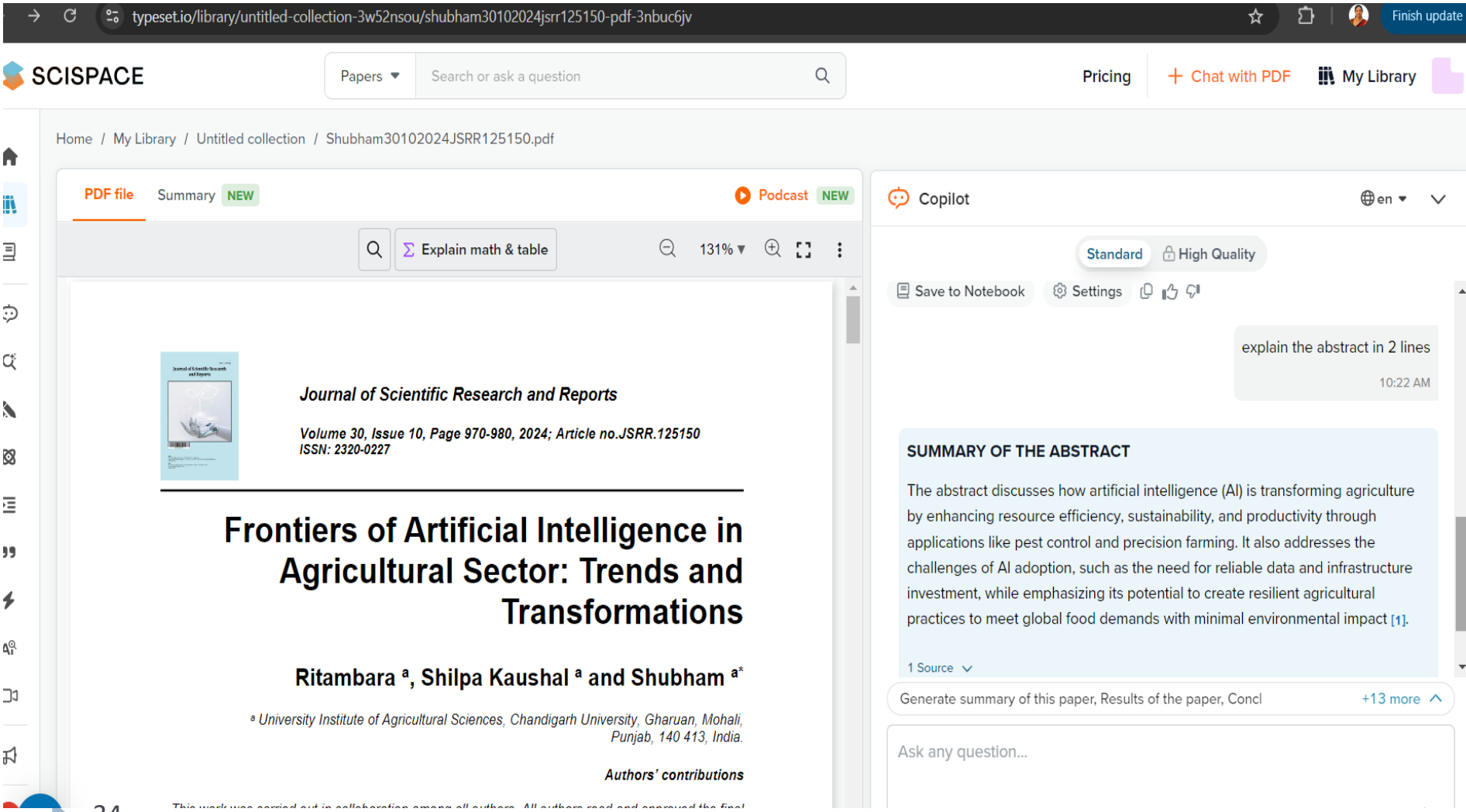
Thank You

Typeset

Citations


SCISPACE
Papers
AI in agriculture
Pricing
Chat with PDF
My I
My Searches / AI in agriculture
Save
High Quality
en
Answer from top 5 papers
AI is revolutionizing agriculture by enhancing productivity, optimizing resource use, and promoting sustainability through precision farming techniques. The integration of AI technologies enables farmers to make data-driven decisions, improving crop management and reducing environmental impacts. Key applications include crop health monitoring, predictive analytics, and automated pest detection, which collectively contribute to increased efficiency and yield.
Key Applications of AI in Agriculture
Precision Farming: AI systems utilize remote sensing and data analytics to monitor soil health and crop conditions, leading to tailored fertilization and irrigation strategies (Jeyalakshmi et al., 2024) (Berna et al., 2024).
Resource Optimization: Smart irrigation systems and variable rate applications of chemicals help reduce water and fertilizer usage by up to 30% (Gangwani, 2024).
Automated Solutions: AI-driven robotics enhance harvesting and sorting processes, while machine learning algorithms improve pest and disease detection (Basa, 2024) (Uma et al., 2024).
Challenges and Future Directions
Implementation Barriers: High costs, lack of technical expertise, and data quality issues hinder widespread AI
Read More
Save to Notebook
APA, Bullets
Find Topics

Library and Chat with PDF options



Academic AI Detector

Catch GPT-4, ChatGPT, Jasper, and any AI in scholarly content.

Input Text | Upload PDF

Examples | Chat GPT | Quillbot | Jasper | AI + Human text | Abstract by AI

My text Scans (19)

Artificial intelligence (AI) in agriculture is transforming the sector by improving resources efficiency, sustainability and productivity. Our study examined a number of AI-related applications such as pest control, crop monitoring, precision farming and soil health evaluation. AI powered devices enables automated fertilization, harvesting and irrigation, and therefore, cutting down the labor expenses and resource waste. Predictive analytics in AI helps with crop yield and weather forecasts which ultimately improves the planning and risk management. The paper also discusses the challenges and limitations of AI adoption in agriculture, such as the need for reliable data, technical expertise and infrastructure investment. Ultimately, the findings highlights the AI can have positive transformative potential in creating resilient agricultural practices that can meet the demands of a growing global population while minimizing environmental impact.

AI detection report

33%

Likely AI

The text content has little or no AI

2 out of 6 sentences were AI written.

AI USAGE PROBABILITY

High AI

Moderate AI